

BRAF fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes BRAF gene fusions from the msk_impact_50k_2026 study, covering 179 fusion events. 151/179 fusions (84.4%) were in-frame. 163/179 fusions (91.1%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ($p=0.0133676$, statistically significant at $\alpha=0.05$). Compared to the literature benchmark (PMC5461196: 100.0% in-frame, 100.0% domain-retained), this run observed 84.4% in-frame and 91.1% domain retention.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 179 analyzed fusion events, identifying 67 distinct fusion partner genes. The most frequent partner was KIAA1549, observed in 43 events.

Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 142/151 (94.0%) of in-frame fusions retained the domain, $p=0.0133676$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.04995.

Domain disruption

Disruption of the configured disruption-required domain(s) was tested with Fisher's exact test comparing in-frame fusions against all others; 126/151 (83.4%) of in-frame fusions disrupted the domain, $p=0.318252$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.673267.

Cutpoint detection

Cutpoint detection scanned 178 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 439 aa (permutation-corrected $p=0.00990099$, statistically significant at $\alpha=0.05$). This is 19 aa from the nearest configured domain boundary at 458 aa.

Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: Algorithm failed: ValueError: params['group_field'] is required

Composite evidence score

Composite evidence score reported: confidence_certainty sub-score excluded: no applicable confidence_stats MLE/CI result was supplied.

Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

Joint partner

Joint partner reported: Algorithm failed: ValueError: joint_partner requires a GeneConfig with gene_pair

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
KIAA1549	43	0.24022346368715083	0.13014640731432997	0.017181246107640623	0.8884995221408092	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2681275760859747	1
SND1	17	0.09497206703910614	0.13014640731432997	0.017181246107640623	0.8642224012892828	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.21566425706137207	2
SCRIB	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	1.0	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2084986985597132	3
SVOPL	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	1.0	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2084986985597132	4
TRIM33	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	1.0	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2084986985597132	5
EPB41	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.997716894977169	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20811818105590804	6
PJA2	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.997716894977169	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20811818105590804	7
RBM33	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.997716894977169	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20811818105590804	8
TRIM24	8	0.0446927374301676	0.13014640731432997	0.017181246107640623	0.886986301369863	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2026984638718226	9
MKRN1	10	0.055865921787709494	0.13014640731432997	0.017181246107640623	0.8447488584474886	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1993832848372742	10
CDK5RAP2	4	0.0223463687150838	0.13014640731432997	0.017181246107640623	0.906392694063927	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19848407308247198	11

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ATF7	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19099489338467512	12
DOCK4	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19099489338467512	13
GIPC2	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19099489338467512	14
GTF2I	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19099489338467512	15
KCND2	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19099489338467512	16
NFIA	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19099489338467512	17
OSBPL9	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19099489338467512	18
PRKAR2B	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19099489338467512	19
RRBP1	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19099489338467512	20
SETBP1	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19099489338467512	21
SLC45A3	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19099489338467512	22
SPOCK1	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19099489338467512	23
ZC3HAV1	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19099489338467512	24

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ZNF207	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19099489338467512	25
NRF1	2	0.0111731843575419	0.13014640731432997	0.017181246107640623	0.8812785388127854	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19057398575476775	26
EXOC4	2	0.0111731843575419	0.13014640731432997	0.017181246107640623	0.8675799086757991	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1882908807319367	27
AKAP9	2	0.0111731843575419	0.13014640731432997	0.017181246107640623	0.8664383561643836	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1881006219800341	28
MAD1L1	2	0.0111731843575419	0.13014640731432997	0.017181246107640623	0.865296803652968	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1879103632281315	29
CCDC6	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.8675799086757991	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18642868333901305	30
CMAS	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.8675799086757991	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18642868333901305	31
ERC1	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.8675799086757991	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18642868333901305	32
MINDY4	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.8675799086757991	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18642868333901305	33
PHTF2	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.8675799086757991	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18642868333901305	34
SNX8	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.8675799086757991	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18642868333901305	35
ABCC1	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.865296803652968	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18604816583520786	36
CAPZA2	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.865296803652968	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18604816583520786	37

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
OSBPL7	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.865296803652968	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18604816583520786	38
PRIM2	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.865296803652968	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18604816583520786	39
PRKAR1B	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.865296803652968	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18604816583520786	40
PTPRZ1	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.865296803652968	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18604816583520786	41
RGS3	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.865296803652968	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18604816583520786	42
CLIP1	2	0.0111731843575419	0.13014640731432997	0.017181246107640623	0.8059360730593608	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.178016908129197	43
TMPRSS2	4	0.0223463687150838	0.13014640731432997	0.017181246107640623	0.7678843226788432	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.17539934451829137	44
ABCC2	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.7442922374429224	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.16588073813353363	45
ITGB5	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.7442922374429224	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.16588073813353363	46
MKLN1	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.7442922374429224	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.16588073813353363	47
NEO1	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.7442922374429224	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.16588073813353363	48
UBE2H	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.7442922374429224	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.16588073813353363	49
WEE2	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.7442922374429224	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.16588073813353363	50

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
AGK	14	0.0782122905027933	0.13014640731432997	0.017181246107640623	0.5808871493803001	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.16285512289777068	51
LUC7L2	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.6529680365296804	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.15066003798132663	52
AGAP3	6	0.0335195530726257	0.13014640731432997	0.017181246107640623	0.4330289193302892	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.123314505412713	53
GTF2IRD1	3	0.01675977653631285	0.13014640731432997	0.017181246107640623	0.4657534246575342	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12318199745514956	54
FAM131B	3	0.01675977653631285	0.13014640731432997	0.017181246107640623	0.2891933028919331	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.09375531049421604	55
CARMIL1	2	0.0111731843575419	0.13014640731432997	0.017181246107640623	0.25114155251141557	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.08555115470453946	56
CLEC2L	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.25114155251141557	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.08368895731161581	57
DBI	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.25114155251141557	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.08368895731161581	58
PLPP1	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.25114155251141557	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.08368895731161581	59
RBMS3	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.25114155251141557	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.08368895731161581	60
TBXAS1	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.25114155251141557	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.08368895731161581	61
CUL1	5	0.0279329608938547	0.13014640731432997	0.017181246107640623	0.17305936073059358	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.07812404825317341	62
KLHL12	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.0	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.04183203189304655	63

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
LRBA	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.0	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.04183203189304655	64
PAK2	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.0	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.04183203189304655	65
PWWP2A	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.0	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.04183203189304655	66
TRA2B	1	0.00558659217877095	0.13014640731432997	0.017181246107640623	0.0	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.04183203189304655	67

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
1	17	146	0	15	None	0.36712402624074814	0.43518719238968984
238	18	145	0	15	None	0.3697885050600375	0.43204659308648946
287	18	145	1	14	1.7379310344827585	1.0	-0.0
303	19	144	1	14	1.8472222222222223	1.0	-0.0
327	32	131	5	10	0.48854961832061067	0.3143387647670962	0.5026020577787302
380	72	91	6	9	1.1868131868131868	0.7933319684125757	0.10054504486238983
381	115	48	7	8	2.738095238095238	0.07931888947474959	1.1006233748798022
393	138	25	7	8	6.308571428571429	0.001556599022521673	2.807823246605722
400	139	24	7	8	6.619047619047619	0.0012299959408670453	2.9100963217817015
438	152	11	8	7	12.090909090909092	0.0001283182416528584	3.891711600164127
439	158	5	9	6	21.066666666666666	4.522271339496731e-05	4.344643383231083
744	158	5	10	5	15.8	0.000419446170360627	3.3773237664571667

domain_disruption tables

frame_domain_contingency_table

126	21
25	6

domain_retention tables

frame_domain_contingency_table

142	21
9	6

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ABCC1	1
ABCC2	1
AGAP3	6
AGK	14
AKAP9	2
ATF7	1
CAPZA2	1
CARMIL1	2
CCDC6	1
CDK5RAP2	4
CLEC2L	1
CLIP1	2
CMAS	1
CUL1	5
DBI	1
DOCK4	1
EPB41	1
ERC1	1
EXOC4	2
FAM131B	3
GIPC2	1
GTF2I	1
GTF2IRD1	3
ITGB5	1

Partner_gene	Event_count
KCND2	1
KIAA1549	43
KLHL12	1
LRBA	1
LUC7L2	1
MAD1L1	2
MINDY4	1
MKLN1	1
MKRN1	10
NEO1	1
NFIA	1
NRF1	2
OSBPL7	1
OSBPL9	1
PAK2	1
PHTF2	1
PJA2	1
PLPP1	1
PRIM2	1
PRKAR1B	1
PRKAR2B	1
PTPRZ1	1
PWWP2A	1
RBM33	1
RBMS3	1
RGS3	1
RRBP1	1
SCRIB	1
SETBP1	1
SLC45A3	1
SND1	17
SNX8	1
SPOCK1	1
SVOPL	1
TBXAS1	1

Partner_gene	Event_count
TMPRSS2	4
TRA2B	1
TRIM24	8
TRIM33	1
UBE2H	1
WEE2	1
ZC3HAV1	1
ZNF207	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-00214 58-T01-IM6-1	P-00214 58-T01-IM6		ABCC1-BRAF	ABCC1	in-frame	thre_e_prime	140493152	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ABCC1 (NM_004996) - BRAF (NM_004333) rearrangement: t(7;16)(q34,p13.11)(chr7:g.140493152::chr16:g.16116105) Protein Fusion: in frame {ABCC1:BRAF}	NA
EVT-P-00346 21-T01-IM6-2	P-00346 21-T01-IM6		ABCC2-BRAF	ABCC2	unknown	thre_e_prime	140495500	8	327	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ABCC2 (NM_000392) - BRAF (NM_004333) rearrangement: t(7;10)(q34;24.2)(chr7:g.140495500::chr10:g.101559038) Protein Fusion: mid-exon {ABCC2:BRAF}	NA
EVT-P-00525 82-T01-IM6-3	P-00525 82-T01-IM6		AGAP3-BRAF	AGAP3	unknown	thre_e_prime	150835371	1	1	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1637:AGAP3_c.1177+1690:BRAFIn v Protein Fusion: mid-exon {AGAP3:BRAF}	NA
EVT-P-00505 74-T02-IM6-4	P-00505 74-T02-IM6		AGAP3-BRAF	AGAP3	in-frame	thre_e_prime	150823608	1	1	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1222-2059:AGAP3_c.1140+900:BRAFIn v Protein Fusion: in frame {AGAP3:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0039799-T01-IM6-5	P-0039799-T01-IM6		AGAP3-BRAF	AGAP3	in-frame	three_prime	150821389	1	1	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1221+416:AGAP3_c.1314+504:BRAFInv Protein Fusion: in frame {AGAP3:BRAF}	NA
EVT-P-0050931-T01-IM6-6	P-0050931-T01-IM6		AGK-BRAF	AGK	in-frame	three_prime	141278652	1	1	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-14294:AGK_c.980+911:BRAFInv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0042835-T03-IM6-7	P-0042835-T03-IM6		AGK-BRAF	AGK	in-frame	three_prime	141279302	1	1	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-13644:AGK_c.980+2167:BRAFInv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0042835-T02-IM6-8	P-0042835-T02-IM6		AGK-BRAF	AGK	in-frame	three_prime	141279302	1	1	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-13644:AGK_c.980+2167:BRAFInv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0050635-T01-IM6-9	P-0050635-T01-IM6		AKAP9-BRAF	AKAP9	in-frame	three_prime	140487501	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AKAP9 (NM_005751) - BRAF (NM_004333) fusion: c.6613-1489:AKAP9_c.1141-117:BRAFInv Protein Fusion: in frame {AKAP9:BRAF}	NA
EVT-P-0030237-T03-IM6-10	P-0030237-T03-IM6		AKAP9-BRAF	AKAP9	in-frame	three_prime	140492357	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AKAP9 (NM_005751) - BRAF (NM_004333) fusion: c.3318+1378:AKAP9_c.1140+1751:BRAFInv Protein Fusion: in frame {AKAP9:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-00238 21-T01-IM6-12	P-00238 21-T01-IM6		ATF7-BRAF	ATF7	in-frame	three_prime	140483233	10	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ATF7 (NM_006856) - BRAF (NM_004333) Rearrangement : t(7;12)(q34;q13.13)(chr7:g.140483233:chr12:g.53913299) Protein Fusion: in frame {ATF7:BRAF}	NA
EVT-P-00078 61-T01-IM5-13	P-00078 61-T01-IM5		BRAF-AGAP3	AGAP3	in-frame	three_prime	140490697	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1326+220:AGAP3_c.1141-3313: BRAFInv Protein fusion: in frame (AGAP3-BRAF)	NA
EVT-P-00215 19-T02-IM6-14	P-00215 19-T02-IM6		BRAF-AGAP3	AGAP3	in-frame	three_prime	140493112	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) Rearrangement : c.1326+142:AGAP3_c.1140+996:BRAFInv Protein Fusion: in frame {AGAP3:BRAF}	NA
EVT-P-00097 57-T01-IM5-15	P-00097 57-T01-IM5		BRAF-AGAP3	AGAP3	in-frame	three_prime	140493871	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1222-2154:AGAP3_c.1140+237: BRAFInv Protein fusion: in frame (AGAP3-BRAF)	NA
EVT-P-00368 86-T01-IM6-16	P-00368 86-T01-IM6		BRAF-AGK	AGK	in-frame	three_prime	140494309	8	327	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement: c.101+10859:AGK_c.981-42:BRAFInv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-00255 61-T03-IM6-17	P-00255 61-T03-IM6		BRAF-AGK	AGK	in-frame	three_prime	140498719	7	327	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1-2 fused to BRAF exons 8-18): c.980+1288:BRAF_c.101+772:AGKInv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-00255 61-T01-IM6-18	P-00255 61-T01-IM6		BRAF-AGK	AGK	in-frame	five_prime	140498874	7	327	True	lost	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - AGK (NM_018238) Rearrangement: c.980+1288:BRAF_c.101+772:AGKInv Protein Fusion: in frame {BRAF:AGK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002427-T01-IM6-19	P-002427-T01-IM6		BRAF-AGK	AGK	in-frame	three_prime	140495070	8	327	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement: c.101+16796:AGK_c.981-803:BR AFinv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-000447-T01-IM5-20	P-000447-T01-IM5		BRAF-AGK	AGK	in-frame	three_prime	140495405	8	327	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1 to 2 and BRAF exons 8 to 18): c.101+10076:AGK_c.981-1138:BR AFinv Protein fusion: in frame (AGK-BRAF)	NA
EVT-P-0023378-T01-IM6-21	P-0023378-T01-IM6		BRAF-AGK	AGK	in-frame	five_prime	140498362	7	327	True	lost	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - AGK (NM_018238) rearrangement: c.980+1800:BRAF_c.101+14574:AGKInv Protein Fusion: in frame {BRAF:AGK}	NA
EVT-P-0001832-T02-IM6-22	P-0001832-T02-IM6		BRAF-AGK	AGK	in-frame	three_prime	140494489	8	327	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exon 1 fused in-frame to BRAF exons 8-18): c.102-572:AGK_c.981-222:BR AFinv Protein fusion: in frame (AGK-BRAF)	NA
EVT-P-0000913-T02-IM5-23	P-0000913-T02-IM5		BRAF-AGK	AGK	in-frame	three_prime	140496386	8	327	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) Fusion (AGK exon 2 fused with BRAF exon 8) : c.101+13932:AGK_c.981-2119:BR AFinv Protein fusion: in frame (AGK-BRAF)	NA
EVT-P-0029531-T01-IM6-24	P-0029531-T01-IM6		BRAF-AGK	AGK	unknown	three_prime	140500233	7	303	False	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1-2 fused to BRAF exons 7-18): c.101+4572:AGK_c.909:BR AFinv Protein Fusion: mid-exon {AGK:BRAF}	NA
EVT-P-0030746-T01-IM6-25	P-0030746-T01-IM6		BRAF-AGK	AGK	in-frame	three_prime	140497533	7	327	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.101+8348:AGK_c.980+2629:BR AFinv Protein Fusion: in frame {AGK:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0010945-T01-IM5-26	P-0010945-T01-IM5		BRAF-AGK	AGK	in-frame	three_prime	140497610	7	327	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement : c.101+12646:AGK_c.980+2552:BR AFInv Protein fusion: in frame (AGK-BRAF)	NA
EVT-P-0049035-T01-IM6-85	P-0049035-T01-IM6		BRAF-CLEC2L	CLEC2L	in-frame	five_prime	139215088	18	767	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			BRAF (NM_004333) - CLEC2L (NM_001080511) rearrangement: c.1177+1549:BR AF_c.191-5932:CLEC2Linv Protein Fusion: in frame {BRAF:CLEC2L}	NA
EVT-P-0004795-T01-IM5-87	P-0004795-T01-IM5		BRAF-CUL1	CUL1	in-frame	three_prime	140493811	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1(NM_003592) - BRAF (NM_004333) Rearrangement : c.790-427:CUL1_c.1140+297:BR AFInv Protein fusion: in frame (CUL1-BRAF)	NA
EVT-P-0022257-T01-IM6-88	P-0022257-T01-IM6		BRAF-EPB41	EPB41	in-frame	three_prime	140482687	10	438	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EPB41 (NM_001166005) - BRAF (NM_004333) fusion (EPB41 exons 1-5 fused in-frame with BRAF exons 11-18): t(1;7)(p35.4;q34)(chr1:g.29424227::chr7:g.140482687) Protein Fusion: in frame {EPB41:BRAF}	NA
EVT-P-0001988-T01-IM3-89	P-0001988-T01-IM3		BRAF-FAM131B	FAM131B	in-frame	three_prime	140489629	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM131B (NM_001031690) - BRAF (NM_004333) Deletion : c.175-103:FAM131B_c.1141-2245:BR AFdel Protein fusion: in frame (FAM131B-BRAF)	NA
EVT-P-0010494-T01-IM5-90	P-0010494-T01-IM5		BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	9	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655) -BRA F (NM_004333) Rearrangement : t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) Protein fusion: in frame (GIPC2-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0054046-T01-IM6-93	P-0054046-T01-IM6		BRAF-GTF2IRD1	GTF2IRD1	in-frame	five_prime	73984431	18	767	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			BRAF (NM_004333) - GTF2IRD1 (NM_005685) rearrangement: c.1178-1749:BR AF_c.2320+10446:GTF2IRD1inv Protein Fusion: in frame {BRAF:GTF2IRD1}	NA
EVT-P-0000828-T03-IM6-94	P-0000828-T03-IM6		BRAF-GTF2IRD1	GTF2IRD1	in-frame	five_prime	73988166	18	767	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			BRAF (NM_004333) - GTF2IRD1 (NM_005685) rearrangement: c.1178-597:BR AF_c.2320+14181:GTF2IRD1inv Protein Fusion: in frame {BRAF:GTF2IRD1}	NA
EVT-P-0015847-T02-IM6-95	P-0015847-T02-IM6		BRAF-ITGB5	ITGB5	out-of-frame	five_prime	140497019	8	327	True	lost	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - ITGB5 (NM_002213) Rearrangement : t(3,7)(q21.2,q34)(chr3:g.124579522::chr7:g.140497019) Protein Fusion: out of frame {BRAF:ITGB5}	NA
EVT-P-0014377-T01-IM6-99	P-0014377-T01-IM6		BRAF-NFIA	NFIA	in-frame	three_prime	140486673	9	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NFIA (NM_001145512) - BRAF (NM_004333) rearrangement: t(1;7)(p31.3;q34)(chr1:g.61836194::chr7:g.140486673) Protein fusion: in frame (NFIA-BRAF)	NA
EVT-P-0010398-T01-IM5-100	P-0010398-T01-IM5		BRAF-OSBPL9	OSBPL9	in-frame	three_prime	140484912	10	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		OSBPL9 (NM_148909) - BRAF (NM_004333) Rearrangement : t(1;7)(p32.3;q34)(chr1:g.52228623::chr7:g.140484912) Protein fusion: in frame (OSBPL9-BRAF)	NA
EVT-P-0003502-T01-IM5-101	P-0003502-T01-IM5		BRAF-PJA2	PJA2	in-frame	five_prime	140482379	10	438	True	lost	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA Protein fusion: in frame (BRAF-PJA2)	NA
EVT-P-0036870-T01-IM6-102	P-0036870-T01-IM6		BRAF-PRIM2	PRIM2	in-frame	three_prime	140493232	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PRIM2 (NM_000947) - BRAF (NM_004333) Rearrangement : t(6;7)(p11.1;q34)(chr6:g.57410815::chr7:g.140493232) Protein Fusion: in frame {PRIM2:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009246-T01-IM5-103	P-0009246-T01-IM5		BRAF-RBM33	RBM33	in-frame	three_prime	140482661	10	438	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RBM33 (NM_053043) - BRAF (NM_004333) rearrangement: c.171+526:RBM33_c.1314+160:BRAFInv Protein fusion: in frame (RBM33-BRAF)	NA
EVT-P-0016313-T01-IM6-104	P-0016313-T01-IM6		BRAF-SLC45A3	SLC45A3	unknown	three_prime	140486608	9	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SLC45A3 (NM_033102) - BRAF (NM_004333) fusion (SLC45A3 exons 1 fused to BRAF exons 10-18): t(1;7)(q32.1;q34)(chr1:g.205638452::chr7:g.140486608) Transcript Fusion {SLC45A3:BRAF}	NA
EVT-P-0064764-T01-IM7-105	P-0064764-T01-IM7		BRAF-SND1	SND1	in-frame	five_prime	127357391	18	767	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			BRAF (NM_004333) - SND1 (NM_014390) fusion: c.1140+819:BRAF_c.1039-3950:SND1inv Protein Fusion: in frame {BRAF:SND1}	NA
EVT-P-0029981-T01-IM6-106	P-0029981-T01-IM6		BRAF-SPOCK1	SPOCK1	in-frame	three_prime	140483862	10	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SPOCK1 (NM_004598) - BRAF (NM_004333) fusion: t(5;7)(q31.2;q34)(chr5:g.136401840::chr7:g.140483862) Protein Fusion: in frame {SPOCK1:BRAF}	NA
EVT-P-0064223-T01-IM7-107	P-0064223-T01-IM7		BRAF-TBXAS1	TBXAS1	out-of-frame	five_prime	139614105	18	767	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			BRAF (NM_004333) - TBXAS1 (NM_001061) rearrangement: c.981-245:BRAF_c.336+2985:TBXAS1inv Protein Fusion: out of frame {BRAF:TBXAS1}	NA
EVT-P-0021469-T01-IM6-108	P-0021469-T01-IM6		BRAF-TRIM33	TRIM33	in-frame	three_prime	140481644	11	439	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM33 (NM_015906) - BRAF(NM_004333) rearrangement: t(1;7)(p13.2;q34)(chr1:g.114963985::chr7:g.140481644) Protein Fusion: in frame {TRIM33:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0042835-T03-IM6-109	P-0042835-T03-IM6		BRAF-WEE2	WEE2	unknown	five_prime	140498912	7	327	True	lost	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - WEE2 (NM_001105558) rearrangement: c.980+1250:BR AF_c.1305:WEE2inv Protein Fusion: mid-exon {BRAF:WEE2}	NA
EVT-P-0015985-T03-IM6-111	P-0015985-T03-IM6		CAPZA2-BRAF	CAPZA2	in-frame	three_prime	140491425	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CAPZA2 (NM_006136) - BRAF (NM_004333) fusion: c.40-938:CAPZA2_c.1140+2683:BR AFinv Protein Fusion: in frame {CAPZA2:BRAF}	NA
EVT-P-0033764-T04-IM6-112	P-0033764-T04-IM6		CARMIL1-BRAF	CARMIL1	in-frame	three_prime	25554753	18	767	True	lost		Protein kinase domain; RAS-binding domain; Cysteine-rich domain		LRRC16A (NM_017640) - BRAF (NM_004333) fusion: t(6;7)(p22.2;q34)(chr6:g.25554753::chr7:g.140491509) Protein Fusion: in frame {LRRC16A:BRAF}	NA
EVT-P-0030796-T01-IM6-113	P-0030796-T01-IM6		CARMIL1-BRAF	CARMIL1	in-frame	three_prime	25580104	18	767	True	lost		Protein kinase domain; RAS-binding domain; Cysteine-rich domain		LRRC16A (NM_017640) - BRAF (NM_004333) fusion (LRRC16A exons 1-29 fused to BRAF exons 9-18): t(6;7)(p22.2;q34)(chr6:g.25580104::chr7:g.140491645) Protein Fusion: in frame {LRRC16A:BRAF}	NA
EVT-P-0005834-T01-IM5-114	P-0005834-T01-IM5		CCDC6-BRAF	CCDC6	in-frame	three_prime	140488401	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CCDC6 (NM_005436) -BRA F (NM_004333) Rearrangement : t(10;7)(chr10q21.2;q34)(chr10:g.61627617::chr7:g.140488401) Protein fusion: in frame (CCDC6-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0038194-T01-IM6-115	P-0038194-T01-IM6		CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140484954	10	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement: t(7;9)(q34;q33.2)(chr7:g.140484954::chr9:g.123187156) Protein Fusion: in frame {CDK5RAP2:BRAF}	NA
EVT-P-0004203-T01-IM5-116	P-0004203-T01-IM5		CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140491585	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement: t(7;9)(7q34;9q33.2)(chr7:140491585::chr9:123251876) Protein fusion: in frame (CDK5RAP2-BRAF)	NA
EVT-P-0005461-T02-IM5-117	P-0005461-T02-IM5		CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140481684	11	439	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement : t(7;9)(7q34;9q33.2)(chr7:140481684::chr9:123255452) Protein fusion: in frame (CDK5RAP2-BRAF)	NA
EVT-P-0014969-T01-IM6-118	P-0014969-T01-IM6		CDK5RAP2-BRAF	CDK5RAP2	in-frame	five_prime	140493700	8	380	True	lost	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - CDK5RAP2 (NM_018249) Rearrangement : t(7;9)(q34;q33.3)(chr7:g.140493700::chr9:g.123269531) Protein fusion: in frame (BRAF-CDK5RAP2)	NA
EVT-P-0040461-T01-IM6-119	P-0040461-T01-IM6		CLIP1-BRAF	CLIP1	in-frame	three_prime	140488218	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CLIP1 (NM_001247997) - BRAF (NM_004333) fusion: t(7;12)(q34;q24.31)(chr7:g.140488218::chr12:g.122833264) Protein Fusion: in frame {CLIP1:BRAF}	NA
EVT-P-0007266-T02-IM6-120	P-0007266-T02-IM6		CLIP1-BRAF	CLIP1	in-frame	three_prime	140495673	8	327	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CLIP1 (NM_001247997) - BRAF (NM_004333) fusion: t(7;12)(q34;q24.31)(chr7:g.140495673::chr12:g.122767026) Protein Fusion: in frame {CLIP1:BRAF}	NA
EVT-P-0024995-T01-IM6-121	P-0024995-T01-IM6		CMAS-BRAF	CMAS	unknown	three_prime	140488046	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		BRAF (NM_004333) rearrangement: t(7;12)(q34;p12.1)(chr7:g.140488046::chr12:g.22218755) Transcript Fusion {CMAS:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-00396 48-T01-IM6-122	P-00396 48-T01-IM6		CUL1-BRAF	CUL1	in-frame	thre_prime	148460538	1	1	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+2950:CUL1_c.1140+1215:BRAFInv Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-00396 48-T03-IM6-123	P-00396 48-T03-IM6		CUL1-BRAF	CUL1	in-frame	thre_prime	148460538	1	1	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+2950:CUL1_c.1140+1215:BRAFInv Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-00640 88-T01-IM7-124	P-00640 88-T01-IM7		CUL1-BRAF	CUL1	in-frame	thre_prime	148459544	1	1	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+1956:CUL1_c.1141-2067:BRAFInv Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-00681 73-T01-IM7-125	P-00681 73-T01-IM7		CUL1-BRAF	CUL1	in-frame	thre_prime	148457779	1	1	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+191:CUL1_c.1140+2118:BRAFInv Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-00501 96-T01-IM6-126	P-00501 96-T01-IM6		DBI-BRAF	DBI	in-frame	thre_prime	120128894	18	767	True	lost		Protein kinase domain; RAS-binding domain; Cysteine-rich domain		DBI (NM_001178017) - BRAF (NM_004333) fusion: t(2;7)(q14.2;q34)(chr2:g.120128894::chr7:g.140486284) Protein Fusion: in frame {DBI:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0064588-T01-IM7-127	P-0064588-T01-IM7		DOCK4-BRAF	DOCK4	out-of-frame	thre_prime	140483872	10	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		DOCK4 (NM_014705) - BRAF (NM_004333) fusion: c.4650+298:DOCK4_c.1178-915:BRAFdup Protein Fusion: out of frame {DOCK4:BRAF}	NA
EVT-P-0014833-T02-IM6-128	P-0014833-T02-IM6		ERC1-BRAF	ERC1	in-frame	thre_prime	140490668	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ERC1 (NM_178040) - BRAF (NM_004333) fusion: t(7;12)(q34;p13.33)(chr7:g.140490668::chr12:g.1246645) Protein Fusion: in frame {ERC1:BRAF}	NA
EVT-P-0036937-T01-IM6-129	P-0036937-T01-IM6		EXOC4-BRAF	EXOC4	out-of-frame	thre_prime	140490241	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EXOC4 (NM_021807) - BRAF (NM_004333) fusion: c.1514+93469:EXOC4_c.1141-2857:BRAFinv Protein Fusion: out of frame {EXOC4:BRAF}	NA
EVT-P-0024206-T01-IM6-130	P-0024206-T01-IM6		EXOC4-BRAF	EXOC4	out-of-frame	thre_prime	140487956	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EXOC4 (NM_021807) - BRAF (NM_004333) rearrangement: c.87-8072:EXOC4_c.1141-572:BRAFinv Protein Fusion: out of frame {EXOC4:BRAF}	NA
EVT-P-0047918-T04-IM7-131	P-0047918-T04-IM7		FAM131B-BRAF	FAM131B	in-frame	thre_prime	143056532	1	1	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			FAM131B (NM_001031690) - BRAF (NM_004333) fusion: c.175-20:FAM131B_c.1141-53:BRAFdel Protein Fusion: in frame {FAM131B:BRAF}	NA
EVT-P-0047918-T02-IM6-132	P-0047918-T02-IM6		FAM131B-BRAF	FAM131B	in-frame	thre_prime	143056532	1	1	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			FAM131B (NM_001031690) - BRAF (NM_004333) fusion: c.175-20:FAM131B_c.1141-53:BRAFdel Protein Fusion: in frame {FAM131B:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-00448 45-T01-IM6-134	P-00448 45-T01-IM6		GTF2I-BRAF	GTF2I	unknown	three_prime	140485631	9	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GTF2I (NM_032999) - BRAF (NM_004333) fusion: c.768:GTF2I_c.1177+1717:BRAFinv Protein Fusion: mid-exon {GTF2I:BRAF}	NA
EVT-P-00008 28-T05-IM6-135	P-00008 28-T05-IM6		GTF2IRD1-BRAF	GTF2IRD1	in-frame	three_prime	140483546	10	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GTF2IRD1 (NM_005685) - BRAF (NM_004333) rearrangement: c.2320+14147:GTF2IRD1_c.1178-589:BRAFinv Protein Fusion: in frame {GTF2IRD1:BRAF}	NA
EVT-P-00467 45-T01-IM6-136	P-00467 45-T01-IM6		KCND2-BRAF	KCND2	out-of-frame	three_prime	140485867	9	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KCND2 (NM_012281) - BRAF (NM_004333) fusion: c.1115+91797:KCND2_c.1177+1481:BRAFinv Protein Fusion: out of frame {KCND2:BRAF}	NA
EVT-P-00569 52-T01-IM6-138	P-00569 52-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493821	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-141:KIAA1549_c.1140+287:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-00442 70-T02-IM6-139	P-00442 70-T02-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491645	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+1560:KIAA1549_c.1140+2463:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-00275 92-T01-IM6-140	P-00275 92-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140492343	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5247+769:KIAA1549_c.1140+1765:BRADFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0055099-T03-IM6-141	P-0055099-T03-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140490663	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+1357:KIAA1549_c.1141-3279:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0058584-T01-IM7-142	P-0058584-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140491493	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+1663:KIAA1549_c.1140+2615:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0066030-T01-IM7-143	P-0066030-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140482785	10	438	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-1172:KIAA1549_c.1314+36:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0035153-T01-IM6-144	P-0035153-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140491093	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon15 fused with BRAF exon9) : c.4930-2046:KIAA1549_c.1140+3015:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0035153-T03-IM6-145	P-0035153-T03-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140491093	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2046:KIAA1549_c.1140+3015:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0049925-T01-IM6-146	P-0049925-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140489497	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+301:KIAA1549_c.1141-2113:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0035844-T03-IM7-147	P-0035844-T03-IM7		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140493285	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4033-1202:KIAA1549_c.1140+823:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0035844-T04-IM7-148	P-0035844-T04-IM7		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140493285	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4033-1202:KIAA1549_c.1140+823:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0017693-T02-IM6-149	P-0017693-T02-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140489537	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion: c.4930-495:KIAA1549_c.1141-2153:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0062963-T01-IM7-150	P-0062963-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140482198	10	438	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-27:KIAA1549_c.1314+623:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0012575-T01-IM5-151	P-0012575-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140490295	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c.4929+793:KIAA1549_c.1141-2911:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0052411-T05-IM7-152	P-0052411-T05-IM7		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140493530	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2296:KIAA1549_c.1140+578:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-00341 17-T01-IM6-153	P-00341 17-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140483812	10	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4346-57:KIAA1549_c.1178-855:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-00524 11-T03-IM6-154	P-00524 11-T03-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140493530	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2296:KIAA1549_c.1140+578:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-00090 97-T01-IM5-155	P-00090 97-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140488817	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain): c.5247+3837:KIAA1549_c.1141-1433:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-00051 21-T02-IM5-156	P-00051 21-T02-IM5		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140491248	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c.4929+2454:KIAA1549_c.1140+2860:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-00648 72-T01-IM7-157	P-00648 72-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140489277	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4032+4347:KIAA1549_c.1141-1893:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-00137 77-T01-IM5-158	P-00137 77-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140481674	11	439	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549(NM_001164665) - BRAF (NM_004333) rearrangement: c.5248-2734:KIAA1549_c.1315-181:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0040760-T01-IM6-159	P-0040760-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140481540	11	439	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+2627:KIAA1549_c.1315-47:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0014191-T01-IM6-160	P-0014191-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140492679	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion : c.5248-320: KIAA1549_c.1140+1429:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0014381-T01-IM6-161	P-0014381-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140492888	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused to BRAF exon 9 -18): c.4930-2082: KIAA1549_c.1140+1220:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0028611-T01-IM6-162	P-0028611-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140482786	10	438	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+35:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0015540-T01-IM6-163	P-0015540-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140489119	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain): c.4930-69:KIAA1549_c.1141-1735:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0062070-T01-IM7-164	P-0062070-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140490443	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-1064:KIAA1549_c.1141-3059:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-00224 04-T01-IM6-165	P-00224 04-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140482177	10	438	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused with BRAF exons 11-18) : c.5248-2270:KIAA1549_c.1314+644: BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-00040 32-T01-IM5-166	P-00040 32-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140487892	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon 16 fused to BRAF exon 9) : c.5248-1949:KIAA1549_c.1141-508:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-00433 72-T01-IM6-167	P-00433 72-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140487523	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+2204:KIAA1549_c.1141-139:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-00286 11-T02-IM6-168	P-00286 11-T02-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140482786	10	438	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+35:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-00438 74-T01-IM6-169	P-00438 74-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140490001	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-2907:KIAA1549_c.1141-2617:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-00690 24-T04-IM7-170	P-00690 24-T04-IM7		KIAA1549-BRAF	KIAA1549	unknown	thre_prime	140492117	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5059:KIAA1549_c.1140+1991:BR AFdup Protein Fusion: mid-exon {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-00293 61-T01-IM6-171	P-00293 61-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140489625	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion: c.4930-2667:KIAA1549_c.1141-2241:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-00165 57-T01-IM6-172	P-00165 57-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140488576	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in-frame to BRAF exons 9-18) : c.4929+2955:KIAA1549_c.1141-1192:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-00329 82-T01-IM6-173	P-00329 82-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140491506	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5248-10:KIAA1549_c.1140+2602:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-00643 10-T04-IM7-174	P-00643 10-T04-IM7		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140493446	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+3560:KIAA1549_c.1140+662:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-00211 27-T01-IM6-175	P-00211 27-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140491741	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-10 fused with BRAF exons 9-18) : c.4032+3996:KIAA1549_c.1140+2367:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-00176 93-T01-IM6-176	P-00176 93-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140489537	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (c.1141-2153): c.4930-495:KIAA1549_c.1141-2153:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020125-T01-IM6-177	P-0020125-T01-IM6		KIAA1549-BRAF	KIAA1549	unknown	thre_prime	140492175	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-17 fused to BRAF exons 9-18): c.5260:KIAA1549_c.1140+1933:BRAFdup Protein Fusion: mid-exon {KIAA1549:BRAF}	NA
EVT-P-0045683-T02-IM6-178	P-0045683-T02-IM6		KIAA1549-BRAF	KIAA1549	unknown	thre_prime	140492850	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5189:KIAA1549_c.1140+1258:BRADFdup Protein Fusion: mid-exon {KIAA1549:BRAF}	NA
EVT-P-0019127-T01-IM6-179	P-0019127-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140490657	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 9-18): c.5247+2110:KIAA1549_c.1141-3273:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0066022-T01-IM7-180	P-0066022-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	thre_prime	140488035	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-4074:KIAA1549_c.1141-651:BRADFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0036680-T01-IM6-181	P-0036680-T01-IM6		KLHL12-BRAF	KLHL12	in-frame	thre_prime	202880970	1	1	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			KLHL12 (NM_021633) - BRAF (NM_004333) fusion (KLHL12 exons 1-4 fused to BRAF exons 9-18): t(1;7)(q32.1;q34)(chr1:g.202880970::chr7:g.140493984) Protein Fusion: in frame {KLHL12:BRAF}	NA
EVT-P-0047833-T02-IM6-183	P-0047833-T02-IM6		LRBA-BRAF	LRBA	in-frame	thre_prime	151502269	1	1	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			LRBA (NM_001199282) - BRAF (NM_004333) fusion: t(4;7)(q31.3;q34)(chr4:g.151502269::chr7:g.140493444) Protein Fusion: in frame {LRBA:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-00064 12-T01-IM5-184	P-00064 12-T01-IM5		LUC7L2-BRAF	LUC7L2	unknown	five_prime	140501109	6	287	True	lost	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		LUC7L2 (NM_001244584) - BRAF (NM_004333) Rearrangement : c.1008-2090:LUC7L2_c.860+103:BRAFdel Antisense fusion	NA
EVT-P-00261 58-T04-IM6-185	P-00261 58-T04-IM6		MAD1L1-BRAF	MAD1L1	in-frame	three_prime	140491749	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MAD1L1 (NM_001013837) - BRAF (NM_004333) fusion: c.1417-9863:MAD1L1_c.1140+2359:BRAFdup Protein Fusion: in frame {MAD1L1:BRAF}	NA
EVT-P-00261 58-T01-IM6-186	P-00261 58-T01-IM6		MAD1L1-BRAF	MAD1L1	in-frame	three_prime	140491749	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MAD1L1 (NM_001013837) - BRAF (NM_004333) Rearrangement : c.1417-9863_c.1140+2359dup Protein Fusion: in frame {MAD1L1:BRAF}	NA
EVT-P-00066 01-T03-IM6-187	P-00066 01-T03-IM6		MINDY4-BRAF	MINDY4	in-frame	three_prime	140490696	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM188B (NM_032222) - BRAF(NM_004333) Fusion (FAM188B exon5 fused with BRAF exon9) : c.664-366:FAM188B_c.1141-3312:BRAF Protein Fusion: in frame {FAM188B:BRAF}	NA
EVT-P-00232 20-T01-IM6-188	P-00232 20-T01-IM6		MKLN1-BRAF	MKLN1	in-frame	three_prime	140496796	8	327	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKLN1 (NM_013255) - BRAF (NM_004333) rearrangement: c.98+4070:MKLN1_c.981-2529:BRAFinv Protein Fusion: in frame {MKLN1:BRAF}	NA
EVT-P-00353 93-T02-IM6-189	P-00353 93-T02-IM6		MKRN1-BRAF	MKRN1	in-frame	three_prime	140488436	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.771+833:MKRN1_c.1141-1052:BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA
EVT-P-00090 24-T01-IM5-190	P-00090 24-T01-IM5		MKRN1-BRAF	MKRN1	in-frame	three_prime	140493485	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.772-184G:MKRN1_c.1140+623:BRAFdup Protein fusion: in frame (MKRN1-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008083-T03-IM6-191	P-0008083-T03-IM6		MKRN1-BRAF	MKRN1	out-of-frame	threshold	140484511	10	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.315-432:MKRN1_c.1178-1554:BRAFdup Protein Fusion: out of frame {MKRN1:BRAF}	NA
EVT-P-0001036-T01-IM3-192	P-0001036-T01-IM3		MKRN1-BRAF	MKRN1	in-frame	threshold	140482250	10	438	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA Protein fusion: in frame (MKRN1-BRAF)	NA
EVT-P-0047135-T01-IM6-193	P-0047135-T01-IM6		MKRN1-BRAF	MKRN1	in-frame	threshold	140492510	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.772-972:MKRN1_c.1140+1598:BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA
EVT-P-0004373-T02-IM5-194	P-0004373-T02-IM5		MKRN1-BRAF	MKRN1	unknown	threshold	140434466	18	744	False	lost		Protein kinase domain; RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.314+4351:MKRN1_c.2232:BRAFdup Protein fusion: mid-exon (MKRN1-BRAF)	NA
EVT-P-0009024-T02-IM6-195	P-0009024-T02-IM6		MKRN1-BRAF	MKRN1	in-frame	threshold	140493485	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.772-185:MKRN1_c.1140+623:BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA
EVT-P-0031896-T02-IM6-196	P-0031896-T02-IM6		MKRN1-BRAF	MKRN1	in-frame	threshold	140482581	10	438	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion (MKRN1 exons 1-4 fused to BRAF exons 11-18): c.772-1018:MKRN1_c.1314+240:BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008083-T02-IM6-197	P-0008083-T02-IM6		MKRN1-BRAF	MKRN1	out-of-frame	thre_prime	140484505	10	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion (MKRN1 exons 1-2 fused with BRAF exons 10-18): c.315-428:MKRN1_c.1178-1548:BRAFdup Protein Fusion: out of frame {MKRN1:BRAF}	NA
EVT-P-0013536-T01-IM5-198	P-0013536-T01-IM5		MKRN1-BRAF	MKRN1	in-frame	thre_prime	140483618	10	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.544+172:MKRN1_c.1178-661:BRAFdup Protein fusion: in frame (MKRN1-BRAF)	NA
EVT-P-0048036-T01-IM6-199	P-0048036-T01-IM6		NEO1-BRAF	NEO1	in-frame	thre_prime	140497651	7	327	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NEO1 (NM_002499) - BRAF (NM_004333) fusion: t(7;15)(q34;q24.1)(chr7:g.140497651::chr15:g.73575682) Protein Fusion: in frame {NEO1:BRAF}	NA
EVT-P-0043972-T03-IM6-200	P-0043972-T03-IM6		NRF1-BRAF	NRF1	in-frame	thre_prime	140487712	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NRF1 (NM_005011) - BRAF (NM_004333) fusion: c.607-7410:NRF1_c.1141-328:BRAFInv Protein Fusion: in frame {NRF1:BRAF}	NA
EVT-P-0065446-T01-IM7-201	P-0065446-T01-IM7		NRF1-BRAF	NRF1	in-frame	thre_prime	140483810	10	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NRF1 (NM_005011) - BRAF (NM_004333) fusion: c.1349-10893:NRF1_c.1178-853:BRAFInv Protein Fusion: in frame {NRF1:BRAF}	NA
EVT-P-0059002-T01-IM7-202	P-0059002-T01-IM7		OSBPL7-BRAF	OSBPL7	in-frame	thre_prime	140491530	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		OSBPL7 (NM_145798) - BRAF (NM_004333) fusion: t(7;17)(q34;q21.32)(chr7:g.140491530::chr17:g.45890381) Protein Fusion: in frame {OSBPL7:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0054958-T02-IM6-203	P-0054958-T02-IM6		PAK2-BRAF	PAK2	in-frame	thre_prime	196530579	1	1	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			PAK2 (NM_002577) - BRAF (NM_004333) rearrangement: t(3;7)(q29;q34)(chr3:g.196530579::chr7:g.140486065) Protein Fusion: in frame {PAK2:BRAF}	NA
EVT-P-0008789-T01-IM5-205	P-0008789-T01-IM5		PHTF2-BRAF	PHTF2	in-frame	thre_prime	140489236	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PHTF2 (NM_001127358) - BRAF (NM_004333) rearrangement: c.2224-367:PHTF2_c.1141-1852:BRAFinv Protein fusion: in frame {PHTF2-BRAF}	NA
EVT-P-0044819-T01-IM6-206	P-0044819-T01-IM6		PLPP1-BRAF	PLPP1	out-of-frame	thre_prime	54789192	18	767	True	lost		Protein kinase domain; RAS-binding domain; Cysteine-rich domain		PPAP2A (NM_176895) - BRAF (NM_004333) fusion: t(5;7)(q11.2;q34)(chr5:g.54789192::chr7:g.140482123) Protein Fusion: out of frame {PPAP2A:BRAF}	NA
EVT-P-0009694-T01-IM5-207	P-0009694-T01-IM5		PRKAR1B-BRAF	PRKAR1B	in-frame	thre_prime	140491855	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PRKAR1B (NM_001164758) -BRAf (NM_004333) Rearrangement : c.892-2551:PRKAR1B_c.1140+2253:BRAFdup Protein fusion: in frame {PRKAR1B-BRAF}	NA
EVT-P-0008567-T01-IM5-208	P-0008567-T01-IM5		PRKAR2B-BRAF	PRKAR2B	in-frame	thre_prime	140486416	9	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PRKAR2B (NM_002736) - BRAF (NM_004333) rearrangement: c.307+8063:PRKAR2B_c.1177+932:BRAFinv Protein fusion: in frame {PRKAR2B-BRAF}	NA
EVT-P-0041181-T01-IM6-209	P-0041181-T01-IM6		PTPRZ1-BRAF	PTPRZ1	in-frame	thre_prime	140492428	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PTPRZ1 (NM_002851) - BRAF (NM_004333) fusion: c.5367+457:PTPRZ1_c.1140+1680:BRAFinv Protein Fusion: in frame {PTPRZ1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-00128 84-T01-IM5-210	P-00128 84-T01-IM5		PWWP2A-BRAF	PWWP2A	in-frame	thre_prime	159539324	1	1	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			PWWP2A (NM_001130864) - BRAF(NM_004333) Rearrangement : t(5;7)(q33.3;q34)(chr5:g.159539324::chr7:g.140495243) Protein fusion: in frame (PWWP2A-BRAF)	NA
EVT-P-00682 60-T01-IM7-211	P-00682 60-T01-IM7		RBMS3-BRAF	RBM S3	in-frame	thre_prime	30030473	18	767	True	lost		Protein kinase domain; RAS-binding domain; Cysteine-rich domain		RBMS3 (NM_001003793) - BRAF (NM_004333) fusion: t(3;7)(p24.1;q34)(chr3:g.30030473::chr7:g.140482707) Protein Fusion: in frame {RBMS3:BRAF}	NA
EVT-P-00647 64-T01-IM7-212	P-00647 64-T01-IM7		RGS3-BRAF	RGS3	out-of-frame	thre_prime	140493282	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RGS3 (NM_144488) - BRAF (NM_004333) fusion: t(7;9)(q34;q32)(chr7:g.140493282::chr9:g.116291333) Protein Fusion: out of frame {RGS3:BRAF}	NA
EVT-P-00413 08-T02-IM6-213	P-00413 08-T02-IM6		RRBP1-BRAF	RRBP1	in-frame	thre_prime	140485917	9	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RRBP1 (NM_004587) - BRAF (NM_004333) fusion: t(7;20)(q34;p12.1)(chr7:g.140485917::chr20:g.17634546) Protein Fusion: in frame {RRBP1:BRAF}	NA
EVT-P-00073 92-T01-IM5-214	P-00073 92-T01-IM5		SCRIB-BRAF	SCRIB	in-frame	thre_prime	140482150	11	439	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SCRIB (NM_182706) - BRAF (NM_004333) rearrangement: t(7;8)(q34;q24.3)(chr7:g.140482150::chr8:g.144876675) Protein fusion: in frame (SCRIB-BRAF)	NA
EVT-P-00566 39-T01-IM6-215	P-00566 39-T01-IM6		SETBP1-BRAF	SETBP1	in-frame	thre_prime	140486594	9	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SETBP1 (NM_015559) - BRAF (NM_004333) fusion: t(7;18)(q34;q12.3)(chr7:g.140486594::chr18:g.42638299) Protein Fusion: in frame {SETBP1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-00489 50-T01-IM6-216	P-00489 50-T01-IM6		SND1-BRAF	SND1	in-frame	thre_e_prime	140492033	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+30037:SND1_c.1140+2075:BRAFInv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-00123 95-T01-IM5-217	P-00123 95-T01-IM5		SND1-BRAF	SND1	in-frame	thre_e_prime	140493527	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1152+27174:SND1_c.1140+581:BRAFInv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-00096 19-T01-IM5-218	P-00096 19-T01-IM5		SND1-BRAF	SND1	in-frame	thre_e_prime	140482494	10	438	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1968+2394:SND1_c.1314+327:BRAFInv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-00522 99-T04-IM6-219	P-00522 99-T04-IM6		SND1-BRAF	SND1	unknown	thre_e_prime	140482935	10	400	False	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1038+2929:SND1_c.1200:BRAFInv Protein Fusion: mid-exon {SND1:BRAF}	NA
EVT-P-00016 87-T01-IM3-220	P-00016 87-T01-IM3		SND1-BRAF	SND1	in-frame	thre_e_prime	140490224	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) inversion: c.1152+28381_c.1141-2840inv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-00041 03-T01-IM5-221	P-00041 03-T01-IM5		SND1-BRAF	SND1	in-frame	thre_e_prime	140481685	11	439	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1038+5799:SND1_c.1315-192:BRAFInv. Protein fusion: in frame (SND1-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-00297 27-T01-IM6-22	P-00297 27-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140482257	10	438	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+12095:SND1_c.1314+564:BRAFInv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-00681 94-T01-IM7-23	P-00681 94-T01-IM7		SND1-BRAF	SND1	in-frame	three_prime	140489175	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+17686:SND1_c.1141-1791:BRAFInv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-00081 40-T01-IM5-24	P-00081 40-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140489439	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-9 fused with BRAF exons 9-18) : c.1039-3849:SND1_c.1141-2055:BRAFInv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-00341 68-T01-IM6-25	P-00341 68-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140489421	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused to BRAF exons 9-18) : c.1152+6972:SND1_c.1141-2037:BRAFInv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-00250 75-T01-IM6-26	P-00250 75-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140490546	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1- 9 with BRAF exons 9-18) : c.1039-2471:SND1_c.1141-3162:BRAFInv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-00020 71-T02-IM5-27	P-00020 71-T02-IM5		SND1-BRAF	SND1	in-frame	three_prime	140489292	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused with BRAF exons 9-18) : c.1152+18606:SN D1_c.1141-1908:BRAFInv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-00021 97-T01-IM3-28	P-00021 97-T01-IM3		SND1-BRAF	SND1	in-frame	five_prime	140488303	9	381	True	lost	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		c.1528-8436:SND1_c.1141-922: BRAFInv Protein fusion: in frame (BRAF-SND1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007533-T01-IM5-229	P-0007533-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140482226	10	438	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 with BRAF exons 11-18) : c.1152+31738:SND1_c.1314+595:BRAFinv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0036567-T01-IM6-230	P-0036567-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140493552	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) Fusion (SND1 exon10 fused with BRAF exon9) : c.1153-22013:SND1_c.1140+556:BRAFinv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0034845-T01-IM6-231	P-0034845-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140489241	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1527+3658:SND1_c.1141-1857:BRAFinv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0054364-T01-IM6-232	P-0054364-T01-IM6		SNX8-BRAF	SNX8	in-frame	three_prime	140488730	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SNX8 (NM_013321) - BRAF (NM_004333) fusion: c.1284+565:SNX8_c.1141-1346:BRAFdup Protein Fusion: in frame {SNX8:BRAF}	NA
EVT-P-0026773-T01-IM6-234	P-0026773-T01-IM6		SVOPL-BRAF	SVOPL	unknown	five_prime	140482097	11	439	True	lost	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) rearrangement: c.1315-604:BRAF_chr7:g.138277426del Transcript Fusion {BRAF:SVOPL}	NA
EVT-P-0051214-T01-IM6-235	P-0051214-T01-IM6		TMPRSS2-BRAF	TMPRSS2	in-frame	three_prime	140502592	6	238	True	retained	Protein kinase domain	RAS-binding domain	Cysteine-rich domain	TMPRSS2 (NM_001135099) - BRAF (NM_004333) fusion: t(7;21)(q34;q22.3)(chr7:g.140502592::chr21:g.42867846) Protein Fusion: in frame {TMPRSS2:BRAF}	NA
EVT-P-0018524-T01-IM6-236	P-0018524-T01-IM6		TMPRSS2-BRAF	TMPRSS2	out-of-frame	three_prime	140490337	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TMPRSS2 (NM_001135099) - BRAF (NM_004333) Rearrangement : t(21;7)(q22.3;q34)(chr21:g.42873801::chr7:g.140490337) Protein Fusion: out of frame {TMPRSS2:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022778-T01-IM6-237	P-0022778-T01-IM6		TMPSRSS2-BRAF	TMPSRSS2	in-frame	thre_prime	140483147	10	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TMPRSS2 (NM_001135099) - BRAF (NM_004333) Rearrangement : t(7;21)(q34;q22.3)(chr7:g.140483147::chr21:g.42878796) Protein Fusion: in frame {TMPRSS2:BRAF}	NA
EVT-P-0043836-T01-IM6-239	P-0043836-T01-IM6		TRA2B-BRAF	TRA2B	in-frame	thre_prime	185650761	1	1	True	retained	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			TRA2B (NM_004593) - BRAF (NM_004333) fusion: t(3;7)(q27.2;q34)(chr3:g.185650761::chr7:g.140491978) Protein Fusion: in frame {TRA2B:BRAF}	NA
EVT-P-0054430-T02-IM6-240	P-0054430-T02-IM6		TRIM24-BRAF	TRIM24	in-frame	thre_prime	140487875	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2719-489:TRIM24_c.1141-491:BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0025699-T02-IM6-241	P-0025699-T02-IM6		TRIM24-BRAF	TRIM24	in-frame	thre_prime	140486854	9	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2014+674:TRIM24_c.1177+494:BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0045111-T01-IM6-242	P-0045111-T01-IM6		TRIM24-BRAF	TRIM24	in-frame	thre_prime	140488049	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2257-760:TRIM24_c.1141-665:BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0066835-T01-IM7-243	P-0066835-T01-IM7		TRIM24-BRAF	TRIM24	in-frame	thre_prime	140490589	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.1879-502:TRIM24_c.1141-3205:BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA

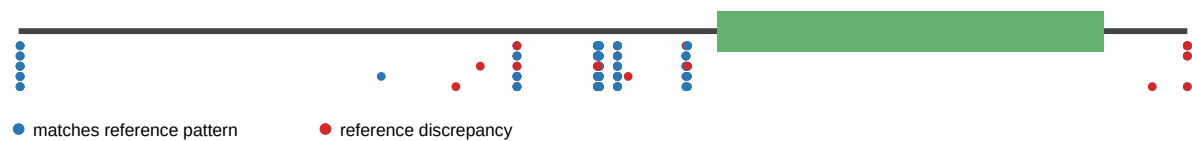
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0068451-T01-IM7-244	P-0068451-T01-IM7		TRIM24-BRAF	TRIM24	in-frame	thre_prime	140482765	10	438	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.484-860:TRIM24_c.1314+56:BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0030988-T01-IM6-245	P-0030988-T01-IM6		TRIM24-BRAF	TRIM24	unknown	thre_prime	140491002	8	380	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2349:TRIM24_c.1140+3106:BRAFInv Protein Fusion: mid-exon {TRIM24:BRAF}	NA
EVT-P-0032569-T01-IM6-246	P-0032569-T01-IM6		TRIM24-BRAF	TRIM24	in-frame	thre_prime	140487936	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) Rearrangement : c.1878+791:TRIM24_c.1141-552inv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0032569-T02-IM6-247	P-0032569-T02-IM6		TRIM24-BRAF	TRIM24	in-frame	thre_prime	140487936	9	381	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.1878+791:TRIM24_c.1141-552:BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0019039-T01-IM6-248	P-0019039-T01-IM6		UBE2H-BRAF	UBE2H	out-of-frame	thre_prime	140499248	7	327	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		BRAF (NM_004333) - UBE2H (NM_003344) rearrangement: c.980+914:BRAF_c.299-4611:UBE2Hdup Protein Fusion: out of frame {UBE2H:BRAF}	NA
EVT-P-0027762-T01-IM6-250	P-0027762-T01-IM6		ZC3HAV1-BRAF	ZC3HAV1	in-frame	thre_prime	140485946	9	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ZC3HAV1 (NM_020119) - BRAF (NM_004333) rearrangement: c.698-431:ZC3HAV1_c.1177+1402:BRAFdup Protein Fusion: in frame {ZC3HAV1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008920-T01-IM5-251	P-0008920-T01-IM5		ZNF207-BRAF	ZNF207	in-frame	thre_prime	140485847	9	393	True	retained	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ZNF207 (NM_001098507) - BRAF (NM_004333) rearrangement: t(7;17)(q34;q11.2)(chr7:g.140485847::chr17:g.30687533) Protein fusion: in frame (ZNF207-BRAF)	NA

Figures

domain retention outliers

BRAF domain-retention track



reference comparison

Reference vs msk_impact_50k_2026

